

Zhihui Tian

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Summary

Ph.D. candidate in Electrical and Computer Engineering at the University of Florida developing scalable machine learning methods for large-scale 3D microstructure evolution. Combines graph neural networks and physics-regularized modeling to improve the efficiency and long-term stability of mesoscale microstructure evolution prediction. Seeking industry research / ML roles and postdoctoral positions.

Education

University of Florida , Electrical and Computer Engineering • Advisor • Prof. Joel B. Harley, SmartDATA Lab. • Research on scalable, physics-guided machine learning for 3D microstructure evolution.	Gainesville, FL May 2023 – present
University of Florida , Electrical and Computer Engineering	Gainesville, FL Aug 2021 – May 2023

Experience

Lawrence Livermore National Laboratory (LLNL) , Collaborative Research Project - Scalable Hybrid CNN-GNN Surrogate for 3D Grain Growth • Designed a hybrid bijective autoencoder + graph neural network surrogate that evolves 3D grain microstructures in a compressed latent space. • Developed a latent-space inference strategy achieving up to 117x memory reduction and 115x runtime improvement on 3D meshes up to 160^3 relative to a GNN-only baseline. • Work published in Acta Materialia (2026) with collaborators at LLNL.	Remote / Gainesville, FL Sept 2025 – present 1 year 1 month
SmartDATA Lab, University of Florida , Graduate Research Assistant • Physics-regulated interpretable ML for 3D microstructure evolution: built a scalable 3D architecture with spatiotemporal extrapolation and inclination/misorientation-dependent grain boundary dynamics; applied it to Lab-DCT experimental data. • Computer vision for experimental microstructure analysis: built a TEM image preprocessing and registration pipeline; fine-tuned U-Net and the Segment Anything Model for microstructure segmentation. • Gaussian-sampled mode filter for grain growth: developed Gaussian-sampled kernels that emulate grain boundary motion, mitigate lattice pinning, and improve adherence to the von Neumann-Mullins law over Monte Carlo Potts models.	Gainesville, FL May 2023 – present 3 years 5 months

Publications

Scaling kinetic Monte-Carlo simulations of grain growth with combined convolutional and graph neural networks

Zhihui Tian, Ethan Suwandi, Tomas Oppelstrup, Vasily V. Bulatov, Joel B. Harley, Fei Zhou
doi.org/10.1016/j.actamat.2026.122153

A mode filter approach to grain growth - improved performance in lattice pinning and von Neumann-Mullins relation

Zhihui Tian, Lin Yang, Kang Yang, Vishal Yadav, Amanda Krause, Michael Tonks, Joel B. Harley

Improved PCA reconstruction-based unsupervised anomaly detection in uncontrolled structural health monitoring with correntropy

Kang Yang, Zhenhan Lin, Zekun Yang, Zhihui Tian, Jie Ma, Jose C. Principe, Joel B. Harley
ieeexplore.ieee.org/document/11078390

Weight decay optimized unsupervised autoencoder-based anomaly detection in uncontrolled dynamic structural health monitoring

Kang Yang, Zekun Yang, Zhihui Tian, Joel B. Harley
doi.org/10.1007/978-3-031-94895-4_3

Quantifying heterogeneous ecosystem services with multi-label soft classification

Zhihui Tian, Jeremy Upchurch, Gabriel A. Simon, Jose Dubeux, Alina Zare, Chang Zhao, Joel B. Harley
doi.org/10.1109/IGARSS53475.2024.10642804

Skills

Machine Learning

Computer Vision

Materials Simulation

Programming and Tools

Languages

Chinese (Mandarin)

Native speaker

English

Fluent